

1 – C:

Key takeaway:

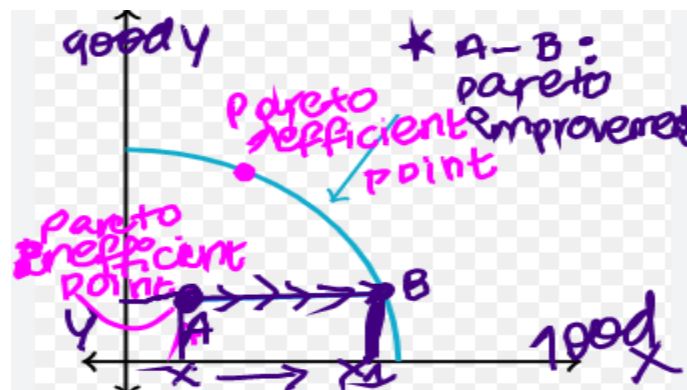
Pareto improvement vs. pareto efficiency:

Pareto Efficiency:

-> A state where no one can be made better off without making someone else worse off. Resources are allocated in the most efficient way possible, and any changes would lead to a loss for at least one person.

Pareto Improvement:

-> A change that makes at least one person better off without making anyone else worse off. This means that resources are reallocated in a way that benefits at least one person, without harming anyone else.



A. This option is incorrect because Pareto improvement only requires that at least one person is made better off, not necessarily both consumers and producers.

B. This option is also incorrect because Pareto efficiency doesn't require equal gains for all parties involved.

C: Pareto improvement is achieved when at least one person is made better off, and no one else is worse off. This means that resources will now be allocated in a way that maximizes overall efficiency, without making anyone worse off.

D. It is allocative efficiency that focuses on maximizing total welfare of the society.

2- D:

External costs, also known as negative externalities, are costs that fall on third parties outside of the direct transaction. In this case, the cost-benefit analysis is for the local council's decision to build a passenger terminal. **The cost to a fishing fleet of relocating to another port is an external cost because it is a consequence of the council's decision that affects a third party (the fishing fleet) who is not directly involved in the transaction.**

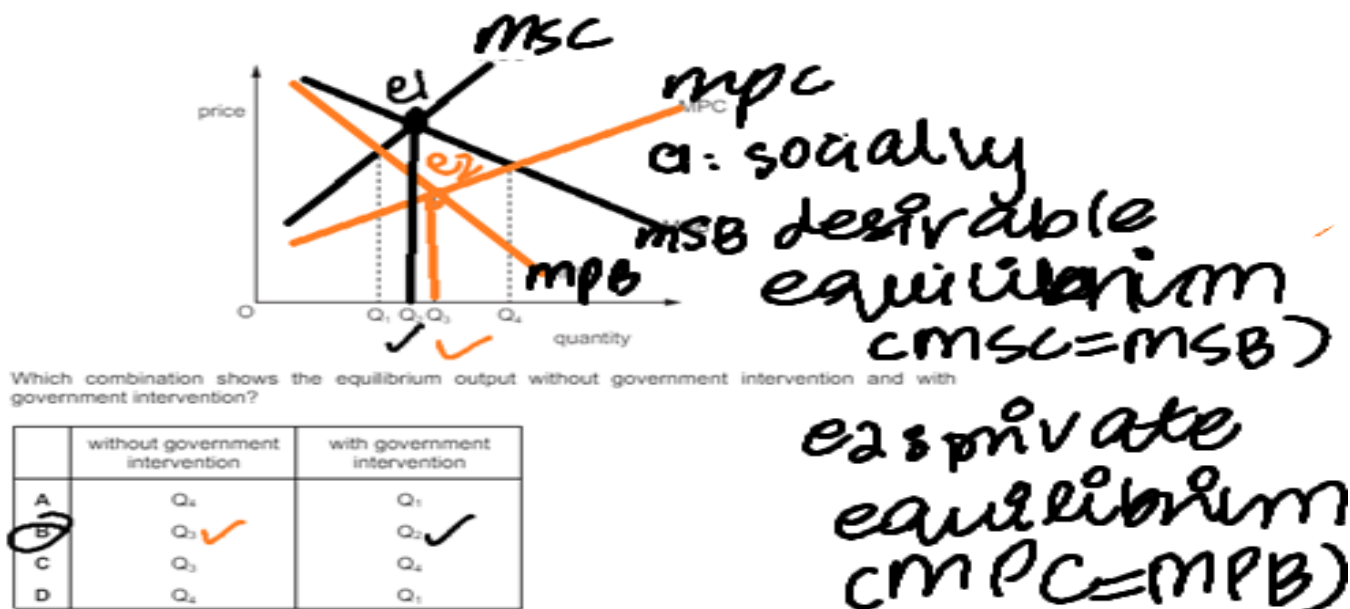
Here's why the other options are incorrect:

A) The cost of a ticket for a passenger to travel on a cruise ship is a private cost, not an external cost.

B) The cost of fuel to operate the cruise ships is a private cost borne by the cruise ship operators.

C) The costs paid by cruise ships when they dock at the port are also private costs.

3-B:



Without government intervention, the private equilibrium where $MPC=MPB$ (shown through "e2") will be operational, and the corresponding equilibrium output Q_3 will be operational

Governments intervene to move the output closer to the socially optimum level of output. This means that the socially desirable level of equilibrium where $MSB=MSC$ (shown through "e1") will be operational. The corresponding level of output will be Q_2 .

This is why option B is correct.

4- C:

1. **Utility maximization:** The consumer wants to equalize the ratio of marginal utility to price for both products, as this ensures they're getting the most utility possible for their money.
2. **Equilibrium condition:** The consumer is in equilibrium when the ratio of marginal utility to price is equal for both products, i.e.:

$$MU_Y / P_Y = MU_Z / P_Z$$

$$MU_Y \times P_Z = MU_Z \times P_Y$$

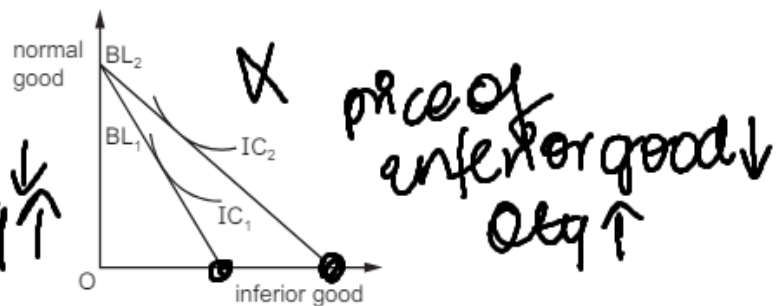
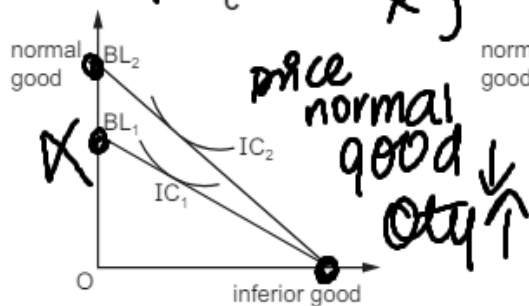
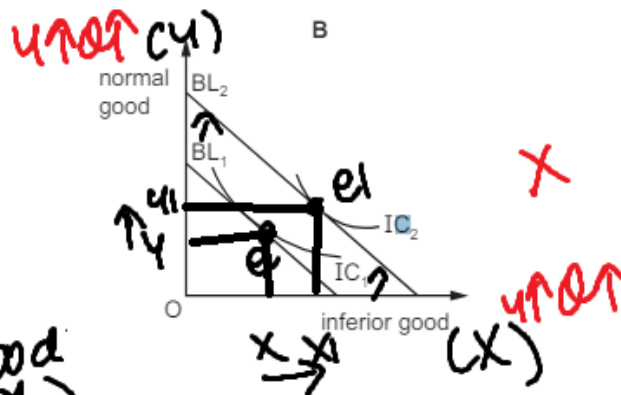
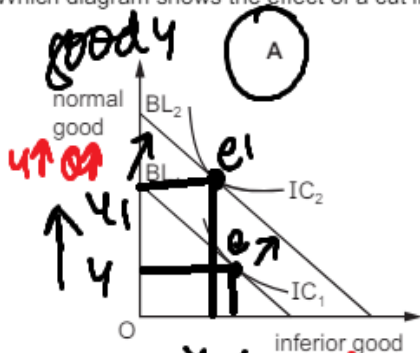
$$\frac{MU_Y}{P_Y} \times P_Z = MU_Z$$

$$\frac{MU_Y}{P_Y} = \frac{MU_Z}{P_Z}$$

5- A:

The diagrams show a consumer's indifference curves (IC) and budget lines (BL) for an inferior good and a normal good.

Which diagram shows the effect of a cut in income tax on the consumer's choice?



Because of a cut in income tax, there is an increase in disposable incomes of individuals, as a result of which, the budget line shifts rightwards from BL1 to BL2.

Moreover, since the new indifference curve is drawn on the opposite direction to the original indifference curve, this represents that the good on the horizontal axis is an inferior good.

Here, an increase in incomes is causing a decrease in the quantity consumed of inferior good from X to X1 since individuals now choose to switch to superior substitutes.

Moreover, this is also causing an increase in the quantity consumed of normal goods from Y to Y1, which represents a positive relationship between income and qty consumed of normal goods.

This explains why option A is correct.

6-A:

Marginal cost refers to the additional cost incurred by producing one additional unit of output or the cost of producing an extra unit of a good or service. It represents the change in total cost resulting from a change in production by a specific quantity.

$$\text{Marginal Cost} = \Delta \text{TC} / \Delta Q$$

* Q_n represents the quantity of output produced
 * Let's assume that the quantity is 10.
 * Since, the marginal cost represents the change in total cost, the formula for marginal cost of 10th unit will be:

$$* MC_{10} = \overset{\text{of producing 10 units}}{TC_{10}} - \overset{\text{of producing 9 units}}{TC_9}$$

* Let's replace it again with n

$$* MC_n = TC_n - TC_{n-1}$$

This explains why key A is correct.

7- C:

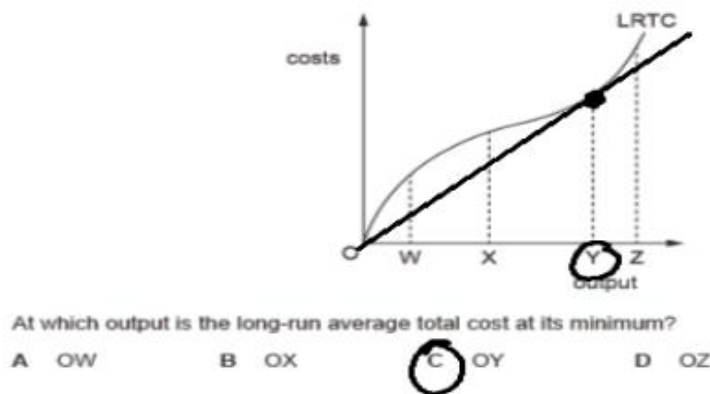
production (tonnes)	total cost (\$)
0	20
1	30
2	35
3	40
4	45
5	50

→ The cost incurred at zero level of output is fixed cost.
 * Fixed cost = 20

* Variable cost = Total cost - Fixed cost
 * $VC = 50 - 20$
 * $VC = 30$

* $AVC = VC / Q_{ty}$
 $AVC = 30 / 5$
 $AVC = 6$

8-C:



Average total cost is at its minimum when the line passing from the origin is tangent to the Total cost curve. This is why, option C is correct.

9-C:

A: This is incorrect because firstly, small firms are more common in services than manufacturing sector, **as they mostly rely on providing personalized services**

Secondly, large firms face “low barriers to exit”, and not high barriers to exit.

B: Although it is correct that small firms are more numerous than large firms, this becomes incorrect because “large firms **do** experience dis-economies of scale”

C: Small firms do well when each item produced is different, that is, when they engage in personalization or customization. And, it is also correct to say that large firms may arise as a result of internal growth or mergers. This is why, this option becomes correct.

D: This is correct to say that small firms cannot have a monopoly power, as a monopoly is a large firm having 25% of the market share. But, it is incorrect to say that large firms cannot earn super-normal profits, as large firms can demonstrate behavior of a monopoly, and may therefore earn super-normal profits.

10-B:

A - This option is incorrect because advertising expenditure is a feature of monopolistic competition, but it doesn't necessarily lead to a difference between short-run and long-run equilibrium. Firms may advertise in both the short and long run.

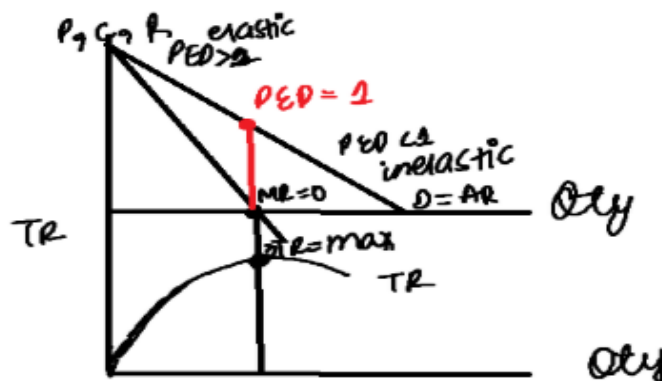
B) Low barriers to entry mean that new firms can easily enter the market in the long-run after having been attracted to short-run super-normal profits. Because of this, an individual demand curve faced by a typical firm shifts to the left and the demand curve becomes more elastic since there is a higher degree of competition now. This results in super-normal profits earned in the short-run to be converted into long-run normal profits.

C - This option is incorrect because diminishing returns affect the firm's cost structure as they increase marginal costs but don't directly impact the difference between short-run and long-run equilibrium.

D – Firms in monopolistic competition do not have monopoly power, as a result of this, this option becomes correct.

11-A:

A – At Q_1P_1 , since marginal revenue is equal to zero, this means that the Total Revenue will be maximized. This can be seen in the diagram below. And, this also explains why, key A, will be correct.



B – Profit will be maximized at the quantity and price when Marginal cost(MC)=Marginal Revenue(MR)

C – Sales maximization represents a situation whereby the firm maximizes the units it sells. This is achieved when the firm charges the lowest possible price it possibly can, or in other words, when normal profits are made(**that is when, Average revenue = Average cost**)

D – Growth maximization occurs when $AR=AC$ (or **Price= Average cost**). At this output and price, the firm is achieving normal profits and it is the highest output consistent with this requirement.

12- A



The 'kinked demand curve' model suggests that the equilibrium price in an oligopoly market will be stable because there is little incentive for firms to change prices.

If for example, the firm raises its price, no other firm in the industry will follow, and therefore, this represents **"elastic demand"** (because, it is the behavior of a firm), and as a result, there is a contraction in quantity demanded by a relatively higher percentage, which results in a "fall in Total Revenue". **This is why, key A, is correct**, because it explains **why a firm would not want to increase its price above "P"**

Importantly, when a firm decreases its price below "P", all other firms in the industry will follow, and therefore, this represents **"inelastic demand"** (because, it is the behavior of an industry), and as a result, there will be an extension in qty demanded by a relatively smaller percentage, which results in a "fall in Total Revenue"

→ **Therefore, prices remain rigid or unchanged at the "kink"**

13-D:

-> Nudge theory, is a concept in behavioral economics that aims to influence people's behavior in a predictable way without limiting their freedom of choice. The essential feature of nudge theory is to use subtle changes in the environment, such as default options or visual cues, to persuade people to make better choices.

- > Nudge theory is based on the idea that people often make suboptimal choices due to cognitive biases and heuristics, and that by designing the environment carefully, policymakers and marketers can "nudge" people toward better decisions.

It believes that the choices of individuals can be nudged or influenced in a particular direction which is in contrast to the view that consumers act in a rational way.

A) Satisficing is a concept in decision-making that refers to accepting a "good enough" option rather than an optimal one.

B) Legal requirements are not a key aspect of nudge theory, which focuses on voluntary choices.

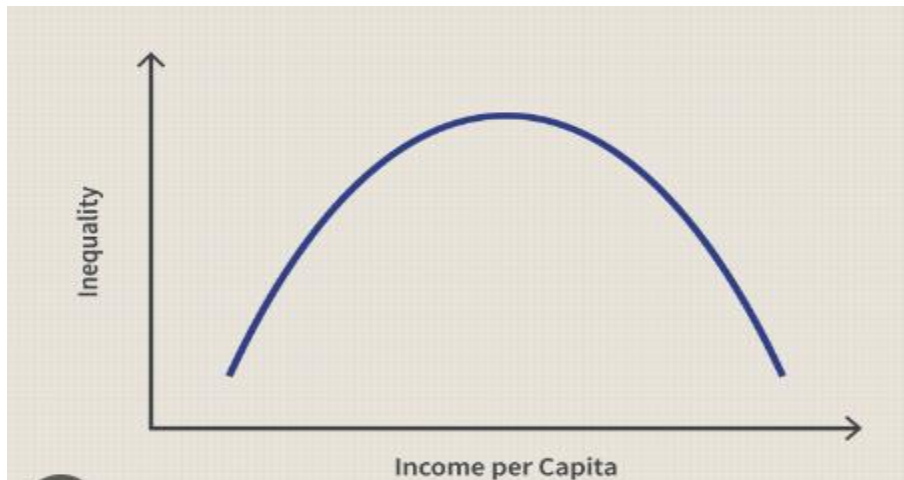
C) Contestable markets refer to a market structure where firms can enter and exit freely.

14- C:

A – A faster rate of economic growth in the early stages of development of an economy translates into a higher level of inequality. This can also be understood through the Kuznets curve, which initially represents a positive relationship between economic growth per capita and gini-coefficient.

This is because initially those with the resources capitalize growth opportunities as growth mainly takes place initially in the manufacturing sector, which means that initially skilled workers benefit.

Additionally, unskilled workers are further put to a disadvantage as a higher level of urbanization or rural to urban migration means that there is a surplus of unskilled workers in the urban areas, as a result of which, the wages of lowly paid workers are further suppressed.



B- A higher human development index represents an improvement in gni per capita, education and health-care, instead of indicating a more equal distribution of income.

C- Gini coefficient is used to understand the level of inequalities in an economy. If an economy has a lower gini coefficient, this represents that **there has been a more equal distribution of income. This explains why key C is correct.**

D – A lower tax as a proportion of gdp represents that the government is going to have less funds available for re-distribution of incomes from the rich to the poor, which will result in a less equal distribution of income.

15- C:

The correct answer is C) Increase in both individual's tax-free allowance for income tax and national minimum wage in real terms.

The poverty trap refers to a situation where low-paid workers are discouraged from working more hours or seeking higher pay due to the fear of losing benefits or paying higher taxes. To reduce the number of low-paid workers caught in the poverty trap, a combination of policies that increases their take-home pay and rewards additional work is needed.

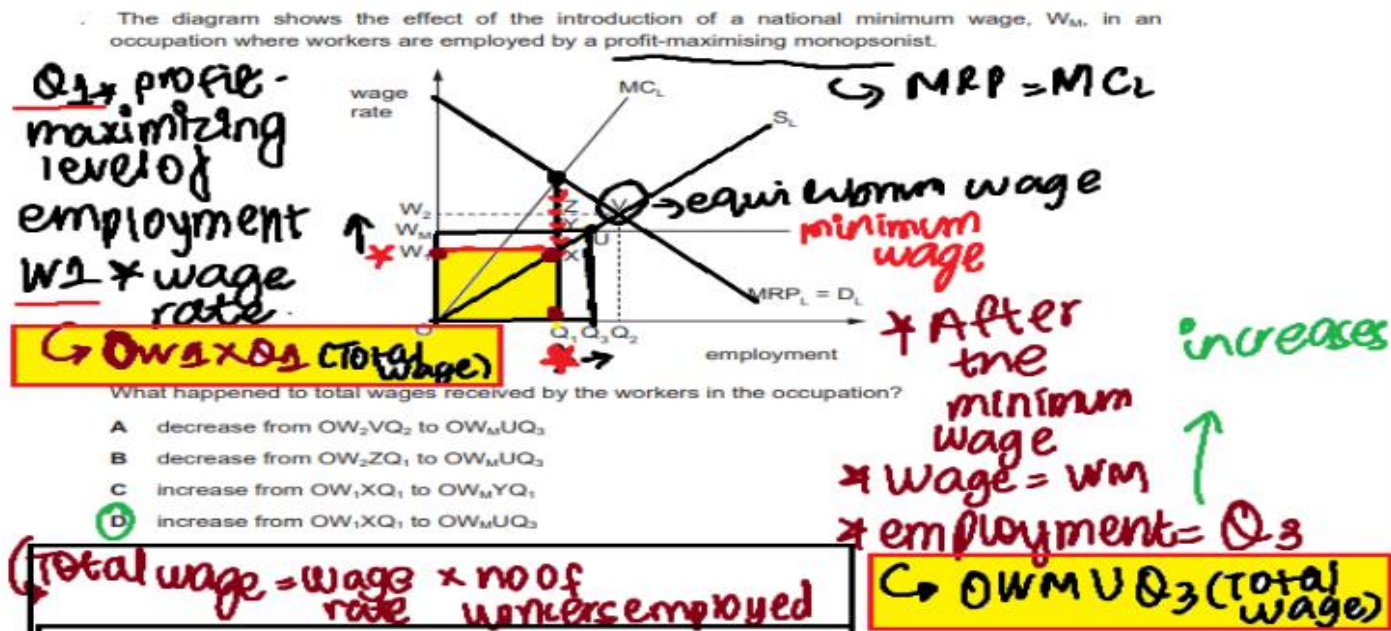
Here's why the combination of policies in option C is most effective:

Tax free allowance represents the amount of income that is exempted from payment of tax. This is most likely to provide tax relief to the low income groups.

- Increasing the individual's tax-free allowance for income tax reduces the amount of tax owed on earnings, allowing low-paid workers to keep more of their hard-earned cash.

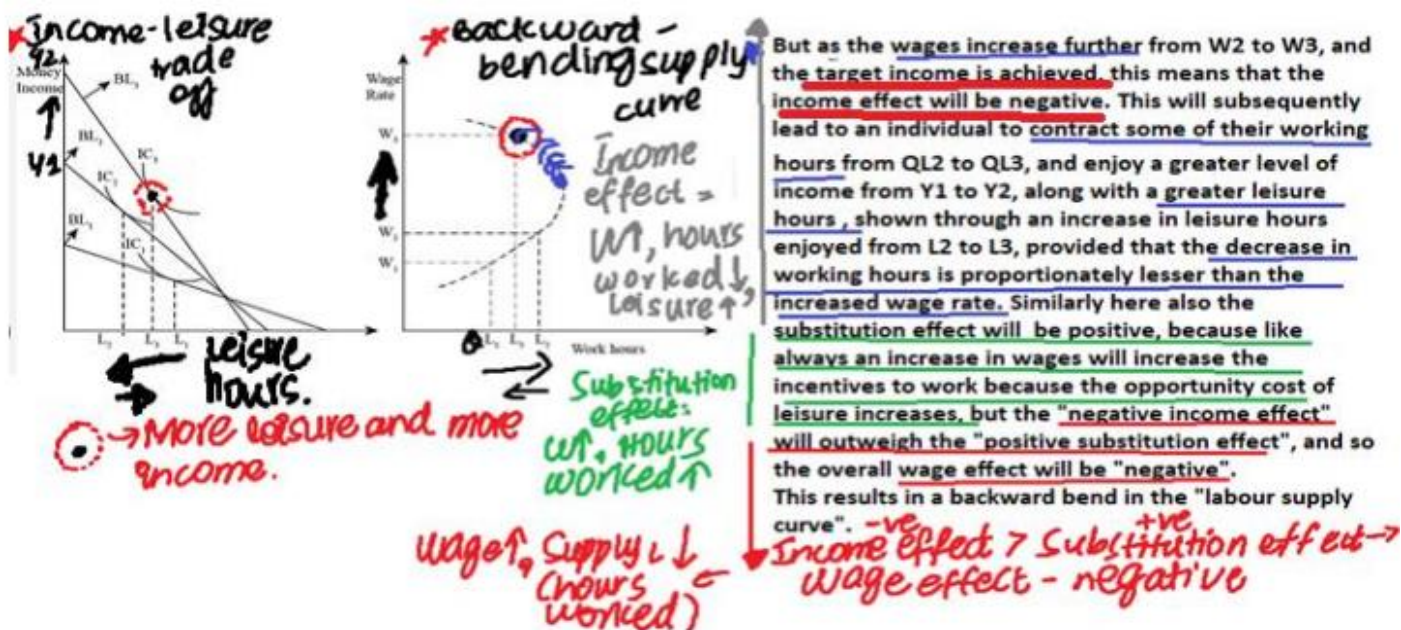
- Increasing the national minimum wage in real terms ensures that low-paid workers receive a higher hourly wage, providing a financial incentive to work more hours or seek better-paying jobs.

16- D:



This explains why, key D, will be correct.

17- C:



This explains why option C is correct as **beyond a certain point, the supply of labor decreases as wages rise.**

18-D:

HDI includes:

- Gni per capita
- Education- mean years of schooling and expected years of schooling
- Health – life expectancy at birth

In the table provided, infant mortality rate and murder rate per 100,000 people are also included, which are not a **part of Human development index.**

This means that we will ignore these two indicators.

The table shows factors that influence living standards.

Which country would be most likely to have the **lowest** Human Development Index (HDI)?

	<i>lowest</i> mean years of schooling	homicide (murder) rate (per 100 000 people)	<i>lowest</i> life expectancy at birth (years)	infant mortality rate (per 100 000 live births)	<i>lowest</i> GNI per capita (PPP\$)
A	13	1	80	4	42 500
B	8	25	75	15	14 500
C	8	3	68	12	18 000
<i>✓</i> D	<i>7</i>	5	<i>58</i>	41	<i>8 400</i>

Country D has the lowest HDI because it has the lowest mean years of schooling, life expectancy at birth and gni per capita.

19- D:

The correct answer is D) Sweden spent the highest proportion of GNI on foreign aid.

To conclude this, we need to calculate the percentage of GNI spent on foreign aid for each country:

- Belgium: $(2.10 / 468.9) \times 100\% \approx 0.45\%$
- Germany: $(23.84 / 3612.1) \times 100\% \approx 0.66\%$
- Sweden: $(5.38 / 532.7) \times 100\% \approx 1.01\%$
- South Korea: $(2.10 / 1403.6) \times 100\% \approx 0.15\%$

Sweden spent the highest proportion of its GNI on foreign aid, at 1.01%.

Here's why the other options are incorrect:

A) While South Korea has the second highest GNI of 1403.6 billion \$, its foreign aid expenditure is very less at about 2.10 billion \$

B) Germany's foreign aid expenditure is not the highest in the world (according to the table); it's only the highest among the four countries listed.

C) South Korea and Belgium did not spend an equal proportion of GNI on foreign aid (0.15% and 0.45%, respectively).

20-C:

The correct answer is C) a fruit picker who cannot find work in the winter months.

Structural unemployment refers to unemployment caused by **a change in the pattern of demand and supply, and mainly occurs when workers cannot respond to this due to geographical and/or occupational immobility.**

The types include:

- **Regional unemployment**
- **Technological unemployment**
- **International unemployment.**

The fruit picker example is a case of seasonal unemployment, which is a type of frictional unemployment. Fruit picking is a seasonal industry, and workers may be temporarily unemployed during the off-season, but they can return to work when the season resumes.

A) The car worker replaced by a robot is an example of technological unemployment which is classed under structural unemployment.

B) The coal miner whose mine closes due to increased use of solar power is an example of industry decline leading to structural unemployment.

D) The textile worker whose factory closes due to production moving abroad is an example of a shift in global trade leading to structural unemployment.

21-D:

When the consumption function shifts upwards from C1 to C2, this represents an increase in autonomous consumption.

A) Higher interest rates make borrowing more expensive, which reduces consumption. When interest rates rise, consumers may stop borrowing for consumption purposes, leading to lower consumption. This would shift the consumption function downwards, not upwards.

B) Higher taxes reduce disposable income, leaving consumers with less money to spend. As a result, consumption decreases.

C) When households save more, they spend less. An increase in savings rate means consumers are setting aside more income, leading to lower consumption at each income level. This would shift the consumption function downwards.

D) If consumers expect sales taxes to increase in the future, they may be incentivized to buy now to avoid higher prices later. As a result, consumption increases in the present, shifting the consumption function upwards. This explains why option D is correct.

22-B:

The accelerator principle is a concept in economics that suggests that investment (I) is linked to changes in the rate of incomes (GDP). **Specifically, it states that investment is a function of the rate of change in income, rather than the level of incomes.** This means that when **the rate of change of income (gdp) increases, demand induced investment will also increase, as now firms increase their demand for capital goods to meet consumer demand.**

The formula to find the coefficient of accelerator is:

Change in Investment/Change in income(GDP)

When changes in interest rate affect investment, the concept of marginal efficiency of investment sets in. This explains why option B becomes correct.

23 – A:

Narrow Money (M1) is a measure of the money supply that includes the most liquid forms of money, such as:

- Physical currency (coins and banknotes)
- Demand deposits (current account deposits)
- Commercial banks' liquid assets held at the central bank

In other words, M1 includes the money that is readily available for transactions. It's called "narrow" because it only includes the most liquid forms of money, excluding near-money assets like savings accounts.

This includes forms of money that are used as a “medium of exchange”

1. Notes and coins in circulation: YES, included in narrow money (M1) as it's the most liquid form of money.
2. Current account deposits: YES, included in M1 as it's money held in checking accounts, readily available for transactions.
3. Banks' liquid assets at the central bank: YES, included in M1 as it's money that banks hold in reserve, readily available for transactions.

4. Treasury bills: NO, not included in M1 as they're not as liquid, having a shorter term but still a form of near-money.

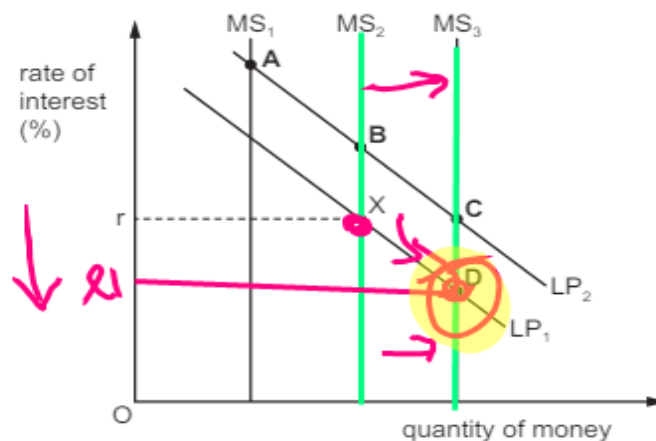
This is why, key A, is correct.

24- D:

If the central bank purchases government securities from the non-bank private sector, the public will receive their money, as a result of which, the money supply increases, shown through a rightward shift from MS_2 to MS_3 . Because of this, **a new equilibrium of "D" will be achieved in the money market.**

The diagram shows three levels of the money supply (MS) and three liquidity preference schedules. The initial money market equilibrium is X. The central bank purchases more government securities.

Which equilibrium shows the initial impact of this purchase?



25 – D:

A) **Incorrect.** Monetarists argue that monetary policy is more effective in reducing inflation, not fiscal policy. Instead, Keynesians argue that fiscal policy is more effective in reducing inflation.

B) **Incorrect.** Monetarists believe that long-term unemployment is caused by structural factors, as according to them, in the long-run, the economy operates at the potential and there is only natural rate of unemployment.

Instead, Keynesians believe that cyclical unemployment persists in the long-run due to wages being inflexible downwards.

C) **Incorrect.** Monetarists assume that the interest elasticity of investment spending is high, meaning that changes in interest rates have a significant impact on investment spending.

D) The qty theory of money is based on the equation: $MV=PT$. Monetarists assume that the velocity of circulation (V) and the volume of transactions (T) are constant, meaning that changes in the money supply (M) directly affect the price level (P) in the long-run. This is why, this is correct.

26- B:

The correct answer is B) Income will increase by an amount greater than the value of autonomous spending.

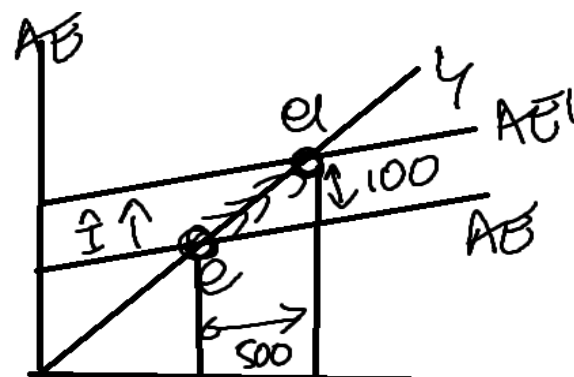
According to Keynesian theory, an increase in autonomous spending (such as government spending or investment) will lead to a multiplier effect, where the initial increase in spending leads to additional rounds of spending, resulting in a larger increase in equilibrium income.

The multiplier effect is calculated as:

Multiplier = $1 / (1 - MPC)$

Where:

MPC = Marginal Propensity to Consume (the proportion of additional income spent on consumption)



* 2 sector economy:-
 * $mpc = 0.8, mps = 0.2$
 Multiplier $\rightarrow \frac{1}{1-0.8} \rightarrow "5"$
 * $\hat{I} \uparrow 100 \rightarrow$ multiplied by 5 $\rightarrow$
 $NI \uparrow 500$

The correct answer is therefore option B.

27- B:

The correct answer is B) Keynesian Monetarist.

Statement 1 "A market economy is inherently unstable and the government must play a stabilizing role through monetary policy and fiscal policy" is a Keynesian view. Keynesians believe that the economy is subject to fluctuations and that government intervention is necessary to stabilize it.

Statement 2 "The economy will always return to its equilibrium rate of output and unemployment and as a result long-run aggregate supply is vertical" is a Monetarist view. Monetarists believe in the concept of the "natural rate of unemployment" and that the economy will always return to this equilibrium rate in the long run, making the long-run aggregate supply curve vertical.

28- B:

- A) Between 2009 and 2010, money GDP did not fall; it increased by approximately 8-9%. Consumer prices also continued to rise, at a rate of 5%.
- B) The graphs show that both money GDP and consumer prices increased in all three years (2008, 2009, and 2010), indicating growth in both nominal GDP and prices. **Athough in some years, money gdp grew at a slower rate, or increased at a decreasing rate, it continued to increase throughout.**
- C) In real terms, GDP did not grow throughout the period. This is because, while money gdp rose by 5% in 2010, consumer prices rose by a higher rate of about 8%, which represents that real gdp may have fallen.
- D) There is no evidence that money GDP fell in any year; it consistently increased. So, we can't conclude that consumer prices rose when money GDP fell, because money GDP never fell.

29-C:

Following depreciation, the prices of imported goods rise, and when their demand is inelastic (**shown through the fact that price elasticity of demand is lesser than 1 (of about 0.3)**), the quantity demanded of imported goods will contract by a relatively smaller percentage and so the import expenditures rise. This will increase the costs of production and reduce Aggregate supply that **will contribute to an increase in cost push inflation.**

Additionally when the export prices fall as a result of depreciation, the quantity demanded will extend by a relatively smaller percentage because the demand is inelastic(ped is 0.3), which will translate into a fall in export revenues, subsequently bringing about a fall in Aggregate demand. **This will lead to a surplus of resources, thereby causing demand pull inflation to decrease.**

This is why key C is correct.

30- B:

A – Contractionary supply side policies reduce Aggregate supply, leading to higher levels of cost push inflation accompanied with a higher degree of unemployment, which is why, this is termed as stagflation. Higher interest rates will dampen investment spending, thereby decreasing Aggregate demand and controlling demand pull inflation, but they will result in an increase in “demand deficient unemployment”

B – Expansionary supply side policies increase Aggregate Supply and reduce cost push inflation while simultaneously reducing unemployment. Additionally, reduced indirect taxes will also decrease costs of production and boost Aggregate supply, helping to achieve **non-inflationary growth**. **This explains why key B will be correct.**

C – Increased direct taxes will reduce disposable income, thereby leading to a fall in consumption and subsequently Aggregate demand. Although this will control demand pull inflation, but it will result in an increase in “demand deficient unemployment”. This is why, this option becomes incorrect.

D – Reduced rates of interest and depreciation of the exchange rate represent “expansionary monetary policy” in action, which is aimed to boost Aggregate demand. Although this reduces demand deficient unemployment, it fails to achieve the objective of price stability as it contributes to demand pull inflation. This is because when an increase in Aggregate demand leads to it outpacing Aggregate supply, there are shortages, which subsequently lead to a rise in general price level.

